

- Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, Jérémy Rapin, Artyom Kozhevnikov, Ivan Evtimov, Joanna Bitton, Manish Bhatt, Cristian Canton Ferrer, Aaron Grattafiori, Wenhan Xiong, Alexandre Défossez, Jade Copet, Faisal Azhar, Hugo Touvron, Louis Martin, Nicolas Usunier, Thomas Scialom, and Gabriel Synnaeve. Code llama: Open foundation models for code, 2024. <https://arxiv.org/abs/2308.12950>.
- Caitlin Sadowski. Simhash : Hash-based similarity detection. 2007. <https://api.semanticscholar.org/CorpusID:199497165>.
- Anshumali Shrivastava and Ping Li. Asymmetric lsh (alsh) for sublinear time maximum inner product search (mips). In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2321–2329, 2014.
- Prajwal Singhania, Siddharth Singh, Shwai He, Soheil Feizi, and Abhinav Bhatele. Loki: Low-rank keys for efficient sparse attention. *arXiv preprint arXiv:2406.02542*, 2024.
- Malcolm Slaney, Yury Lifshits, and Junfeng He. Optimal parameters for locality-sensitive hashing. *Proceedings of the IEEE*, 100(9):2604–2623, 2012. doi: 10.1109/JPROC.2012.2193849.
- Ryan Spring and Anshumali Shrivastava. A new unbiased and efficient class of lsh-based samplers and estimators for partition function computation in log-linear models. *arXiv preprint arXiv:1703.05160*, 2017.
- Hanshi Sun, Zhuoming Chen, Xinyu Yang, Yuandong Tian, and Beidi Chen. Triforce: Lossless acceleration of long sequence generation with hierarchical speculative decoding. *arXiv preprint arXiv:2404.11912*, 2024.
- Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. Quest: Query-aware sparsity for efficient long-context llm inference. *arXiv preprint arXiv:2406.10774*, 2024.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models, 2023.
- Minzheng Wang, Longze Chen, Cheng Fu, Shengyi Liao, Xinghua Zhang, Bingli Wu, Haiyang Yu, Nan Xu, Lei Zhang, Run Luo, et al. Leave no document behind: Benchmarking long-context llms with extended multi-doc qa. *arXiv preprint arXiv:2406.17419*, 2024.
- Wenhao Wu, Yizhong Wang, Guangxuan Xiao, Hao Peng, and Yao Fu. Retrieval head mechanistically explains long-context factuality. *arXiv preprint arXiv:2404.15574*, 2024.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.
- Zihao Ye, Ruihang Lai, Bo-Ru Lu, Chien-Yu Lin, Size Zheng, Lequn Chen, Tianqi Chen, and Luis Ceze. Cascade inference: Memory bandwidth efficient shared prefix batch decoding, February 2024. <https://flashinfer.ai/2024/02/02/cascade-inference.html>.
- Amir Zandieh, Insu Han, Majid Daliri, and Amin Karbasi. Kdeformer: Accelerating transformers via kernel density estimation. In *International Conference on Machine Learning*, pages 40605–40623. PMLR, 2023.
- Hailin Zhang, Xiaodong Ji, Yilin Chen, Fangcheng Fu, Xupeng Miao, Xiaonan Nie, Weipeng Chen, and Bin Cui. Pqcache: Product quantization-based kvcache for long context llm inference. *arXiv preprint arXiv:2407.12820*, 2024.
- Jun Zhang, Jue Wang, Huan Li, Lidan Shou, Ke Chen, Gang Chen, and Sharad Mehrotra. Draft & verify: Lossless large language model acceleration via self-speculative decoding. *CoRR*, abs/2309.08168, 2023a. doi: 10.48550/ARXIV.2309.08168. <https://doi.org/10.48550/arXiv.2309.08168>.

- Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang "Atlas" Wang, and Beidi Chen. H2o: Heavy-hitter oracle for efficient generative inference of large language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 34661–34710. Curran Associates, Inc., 2023b. https://proceedings.neurips.cc/paper_files/paper/2023/file/6ceefa7b15572587b78ecfceb2827f8-Paper-Conference.pdf.
- Yang Zhou. Yangzhoumill/infini_igsm_4k_noise_close. https://huggingface.co/datasets/YangZhoumill/infini_igsm_4k_noise_close, 2024a. Accessed: 2024-10-20.
- Yang Zhou. Yangzhoumill/infini_igsm_8k_noise_close. https://huggingface.co/datasets/YangZhoumill/infini_igsm_8k_noise_close, 2024b. Accessed: 2024-10-20.

Appendix

A Proofs for theorems

A.1 Proof for Theorem 3.2

Proof.

$$\mathbb{E}(\bar{o}) = \frac{1}{\mathcal{B}} \sum_{j=1}^{\mathcal{B}} \mathbb{E}[v_{i_j}] = \frac{1}{\mathcal{B}} \sum_{i=1}^n w_i v_i = o \quad (12)$$

Assume Σ_1 is the covariance matrix of $\bar{o}$, Σ_2 is the covariance matrix of v_i

$$\text{Tr}(\Sigma_1) = \frac{1}{\mathcal{B}} \text{Tr}(\Sigma_2) = \frac{1}{\mathcal{B}} (\mathbb{E}[\|v_i\|^2] - \|\mathbb{E}[v_i]\|^2) = \frac{1}{\mathcal{B}} (\mathbb{E}[\|v_i\|^2] - \|o\|^2) \quad (13)$$

$\mathbb{E}[\|v_X\|^2] - \|o\|^2$ is a constant, so the trace of covariance matrix monotonically decreases with $\mathcal{B}$. $\square$

A.2 Proof for Theorem 3.3

Proof.

$$\mathbb{E}[|S|] = \mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_{i \in S}\right] = \sum_{i=1}^n \mathbb{E}[\mathbf{1}_{i \in S}] = \sum_{i=1}^n (1 - (1 - w_i)^{\mathcal{B}}) = n - \sum_{i=1}^n (1 - w_i)^{\mathcal{B}} \quad (14)$$

Without loss of generality, let $a_i = 1 - w_i$ and $a_1 = \min_{1 \leq i \leq n} a_i = \epsilon$, then

$$\mathbb{E}[|S|] = n - \sum_{i=1}^n a_i^{\mathcal{B}} = n - a_1^{\mathcal{B}} - \sum_{i=2}^n a_i^{\mathcal{B}} \quad (15)$$

$$= n - \epsilon^{\mathcal{B}} - \sum_{i=2}^n a_i^{\mathcal{B}} \quad (16)$$

$f(x) = x^{\mathcal{B}}$ is convex function with $\mathcal{B} \geq 1$ and $x \geq 0$. Then with Jensen's inequality, we have

$$\sum_{i=2}^n a_i^{\mathcal{B}} \geq (n-1) \left(\frac{\sum_{i=2}^n a_i}{n-1} \right)^{\mathcal{B}} = (n-1) \left(\frac{(\sum_{i=1}^n a_i) - a_1}{n-1} \right)^{\mathcal{B}} \quad (17)$$

$$= (n-1) \left(\frac{n-1-\epsilon}{n-1} \right)^{\mathcal{B}} = (n-1) \left(1 - \frac{\epsilon}{n-1} \right)^{\mathcal{B}} \quad (18)$$

Let $g(x) = (1-x)^{\mathcal{B}} + \mathcal{B}x - 1$. We can prove $g(x) \geq 0$ for any $x \in (0, 1)$, $\mathcal{B} \geq 1$. Then we have

$$\sum_{i=2}^n a_i^{\mathcal{B}} \geq (n-1) \left(1 - \frac{\epsilon \mathcal{B}}{n-1} \right) = n-1 - \epsilon \mathcal{B} \quad (19)$$

Then we finally have

$$\mathbb{E}[|S|] = n - \epsilon^{\mathcal{B}} - \sum_{i=2}^n a_i^{\mathcal{B}} \leq 1 + \epsilon \mathcal{B} \quad (20)$$

$\square$

B Oracle sampling

The optimal sampling probability to guarantee estimation is unbiased in terms of lowest variance is not directly using attention score distribution w_i , but $u'_i \propto w_i \|v_i\|$. However, this sampling probability is not optimal in terms of downstream accuracy and efficiency. We attribute this to two reasons. First, we observe the value norm of the sink token is significantly smaller than others (Figure 11), given its lower probability of being sampled, which may influence the functionality of attention. Second, due to the same reason, $u'_i \propto w_i \|v_i\|$ is flatter than w_i , resulting larger computation cost (as analyzed by Theorem 3.3).